TEXORA · COURSE NOTES · LECTURE SHEET

Chemical Equilibrium

Dynamic balance and the law of mass action — K_c and K_p, the reaction quotient, and Le Chatelier's principle at work in industry and the dyebath.

CHE-1105 · CHEMISTRY-I · TOPIC 6 · LEVEL 1 TERM 1

1. A balance that never stops moving

Many reactions do not run to completion; they run until the forward and reverse reactions proceed at **equal rates** and the composition stops changing — **dynamic equilibrium**. Nothing has stopped: molecules keep reacting in both directions, which isotope-tracer experiments prove directly. Macroscopically the state is marked by constant concentrations (in a closed system at fixed temperature), reached from either direction, and shifted only by outside disturbance. Chemical equilibrium is the science of exactly where that balance lies and how to move it — arguably the most economically important idea in this course, because industry is the art of trapping equilibria at useful compositions.

2. The law of mass action and the equilibrium constant

For the general reaction aA + bB ⇌ cC + dD, the law of mass action states that at equilibrium

$$K_c = [C]^c\,[D]^d \,/\, (\,[A]^a\,[B]^b\,)$$

with concentrations in mol L^{-1}. K is a constant at a given temperature only; its size reads like a verdict — $K \gg 1$ means products dominate ("the reaction goes nearly to completion"), $K \ll 1$ means reactants dominate, $K \approx 1$ means a genuine balance. Pure solids and liquids do not appear in the expression (their "concentration" is constant). For gas reactions, partial pressures give K_p, and the two constants are related by

$$K_p = K_c\,(R\,T)^{\Delta n}$$

where Δn is (moles of gaseous products) − (moles of gaseous reactants). If Δn = 0 the constants are equal.

WORKED EXAMPLE — COMPUTING AND CONVERTING K

At equilibrium for $N_2O_4 \rightleftharpoons 2\ NO_2$ in a 1 L vessel, $[N_2O_4] = 0.040$ and $[NO_2] = 0.12$ mol L^{-1}. $K_c = [NO_2]^2 / [N_2O_4] = (0.12)^2 / 0.040 =$ **0.36 mol L^{-1}**. Here $\Delta n = 2 - 1 = +1$, so at 298 K, $K_p = K_c(RT)^1 = 0.36 \times 0.0821 \times 298 \approx$ **8.8 atm**. (Units attach to K whenever $\Delta n \neq 0$ — quote them.)

3. The reaction quotient: which way will it move?

At any moment, the mixture has a **reaction quotient** Q, computed with the same formula as K but from the present concentrations. Comparing Q with K predicts direction without waiting: **Q < K** — too few products, the reaction runs forward; **Q > K** — too many, it runs backward; **Q = K** — already at equilibrium. Equilibrium calculations then become algebra: set up an ICE table (Initial, Change, Equilibrium), substitute into K, and solve — with the standard simplification that when K is small and the starting concentration large, the change x is negligible beside it (always check afterward that x really was under about 5% of the starting value).

4. Le Chatelier’s principle

When a system at equilibrium is disturbed, it shifts in the direction that partially cancels the disturbance. The three classical disturbances:

- **Concentration:** add a reactant and the equilibrium shifts toward products (Q drops below K); remove a product and it likewise chases it. This is why reactions are driven by precipitation, gas escape, or continuous product removal.
- **Pressure (gases only):** compress the mixture and it shifts toward the side with fewer gas moles; expand it and it shifts toward more. Inert gas at constant volume changes nothing (no partial pressure changes); at constant pressure it expands the volume and acts like decompression.
- **Temperature:** the one disturbance that changes K itself. Heat acts as a reactant in an endothermic reaction and as a product in an exothermic one: raising T raises K for endothermic reactions and lowers it for exothermic (the van’t Hoff relation). A **catalyst** changes neither K nor position — it shortens the road to equilibrium equally in both directions, which is why catalysts earn money through speed, not yield.

WORKED EXAMPLE — THE HABER PROCESS AS APPLIED LE CHATELIER

$N_2 + 3\ H_2 \rightleftharpoons 2\ NH_3$, $\Delta H = -92$ kJ mol^{-1}. For yield alone: the product side has 2 gas moles against 4, so **high pressure** pushes right (industry uses 150–250 atm); the reaction is exothermic, so **low temperature** raises K — but low temperature is unbearably slow, so industry compromises near 450 °C with an iron **catalyst** (which speeds both directions equally and changes nothing about position), continuously **removing ammonia** by condensation and recycling unreacted gases.

> Every industrial equilibrium is this same negotiation between where the balance lies and how fast it can be reached.

5. Equilibrium in the dyebath — a textile case

Dyeing is an equilibrium between dye in the bath and dye in the fibre: Dye(bath) ⇌ Dye(fibre). The partition at equilibrium — measured as **exhaustion**, the percentage of dye that has left the bath — is Le Chatelier in work clothes. Raising temperature speeds the approach but, dyeing being slightly exothermic, can lower the final exhaustion; adding electrolyte (for direct and reactive dyes on cotton) shifts the partition toward the fibre by suppressing the fibre surface's negative charge; and reactive dyeing then removes the product from the equilibrium altogether by fixing the dye covalently — the industrial trick of §4, product removal, performed by chemistry. A dyer adjusting salt, temperature and time is solving equilibrium problems whether or not the word is used.

6. Glossary — key terms of this topic

TERM	MEANING
Dynamic equilibrium	The state where forward and reverse reactions proceed at equal rates and composition is constant.
Law of mass action	The principle giving K as the product-to-reactant concentration ratio, each raised to its stoichiometric power.
Reaction quotient (Q)	The mass-action ratio computed at any moment; compared with K it predicts the direction of change.
Le Chatelier's principle	A disturbed equilibrium shifts to partially cancel the disturbance.
ICE table	The Initial–Change–Equilibrium bookkeeping used to solve equilibrium compositions.
Exhaustion	In dyeing, the percentage of dye transferred from bath to fibre at equilibrium.

7. Key points and summary

- Equilibrium is dynamic and quantified by K (K_c from concentrations, K_p from pressures, $K_p = K_c(RT)^{\Delta n}$); Q versus K predicts direction.
- Concentration and pressure shifts move the position without changing K; temperature changes K itself (van't Hoff); catalysts change only the rate of arrival.

- The Haber process and the dyebath show the same negotiating pattern: pressure, temperature, catalyst and product removal balanced against each other.

8. Practice

Answer in your own words; brief model answers follow.

1. For $H_2 + I_2 \rightleftharpoons 2\ HI$ at equilibrium in 1 L: $[H_2] = [I_2] = 0.020$, $[HI] = 0.16$. Compute K_c and state what its size means.
2. For the reaction of question 1, $\Delta n = 0$. What is the relation between K_p and K_c?
3. A mixture for $N_2O_4 \rightleftharpoons 2\ NO_2$ has $[N_2O_4] = 0.10$ and $[NO_2] = 0.10$, with $K_c = 0.36$. Compute Q and predict the direction of change.
4. For the Haber equilibrium, state the effect on ammonia yield of (a) doubling the pressure, (b) raising the temperature, (c) adding a catalyst, (d) condensing out the ammonia as it forms.
5. $PCl_5 \rightleftharpoons PCl_3 + Cl_2$ is endothermic. How does heating affect the degree of dissociation of PCl_5, and why?
6. An equilibrium mixture is compressed and shifts right. What can you infer about the gas mole counts on the two sides?
7. In reactive dyeing, why does covalent fixation of dye onto cotton raise the effective exhaustion beyond the bath⇌fibre equilibrium value?
8. “A catalyst increases the yield of an equilibrium reaction.” True or false — correct the statement.

Answer key

1. $K_c = (0.16)^2 / (0.020 \times 0.020) = 0.0256 / 0.0004 = 64$ — large: the equilibrium lies well toward HI.
2. $K_p = K_c$ exactly ($\Delta n = 0$, so $(RT)^0 = 1$).
3. $Q = (0.10)^2 / 0.10 = 0.10 < K = 0.36$, so the reaction runs forward — more NO_2 forms.
4. (a) Increases (fewer gas moles on the product side). (b) Decreases (exothermic; K falls). (c) No change in yield — faster arrival only. (d) Increases (product removal pulls the equilibrium forward).
5. It increases: for an endothermic reaction heat is effectively a reactant, so heating shifts the equilibrium toward dissociation products (K rises).
6. The product side has fewer moles of gas than the reactant side (compression favours the smaller-volume side).
7. Fixation removes dye from the reversible partition (product removal): bath dye keeps transferring to replace what the fibre has bound, so total uptake exceeds the unfixed equilibrium.
8. False. A catalyst speeds the forward and reverse reactions equally; the equilibrium position and yield are unchanged — only the time to reach them falls.